

INDUSTRIAL DEVICES (INDIA) PVT. LTD.

Think Chlorine. Think IDI.



Gas Chlorination Systems and Safety Devices

AN ISO 9001:2015
&
ISO 45001 : 2018 CERTIFIED COMPANY



COMPANY PROFILE

www.industrialdevices.in

About us

Industrial Devices (India) Pvt. Ltd. with its manufacturing facility, administrative office, design office and servicing department is based in **New Delhi, India.** We are one of the leading Chlorination System manufacturers in India. Established in 1980, we have supplied over 5000+ chlorination system in India and abroad.

The industries where our systems find their applications are numerous. Basically, wherever a large amount of water needs to be treated at the least cost, we are relevant. Some such domains are Municipal Corporations, Thermal Power Stations, Public Health Departments, Chemical, Paper Pulp Industries, etc. We also regularly export our chlorinators to various countries across the globe like UAE, Egypt, Iraq, Iran, South Africa, Myanmar.

We design, manufacture and commission Chlorination Plants ranging from 100 gms/hr to 1000 + Kgs/hr.

Our products include Vacuum Operated Chlorinators, Steam or Electrically Heated Chlorine

Evaporators/Vaporizers, Chlorination accessories like Roller Supports, Manifold, Lifting Beam, Valves etc. We also manufacture other safety items such as Chlorine Leak Absorption Systems, Chlorine Leak Detectors, Emergency Repair Kits to name a few.

The Directors of the company have each been in the field of Water Treatment through Chlorination for more than **35 years.**



Our Team

The team at **Industrial Devices (India) Pvt. Ltd.** is backed by more than 35 years of experience in Chlorination.

With a wide and diverse customer base, our team of experienced engineers has mastered the entire value chain of chlorination system design, manufacture, supply, erection commissioning and maintenance for small and large capacity Chlorination plants.

Each system requirement differs based on its application and the customer we work with. In our years of experience, we have handled numerous technical situations and successfully come out with solutions that works for our customers, which is what keeps them coming back to us time after time.

Our Vision

To be the best and most preferred provider of Chlorination systems across India by ensuring quality, cost competitiveness, customized design and customer satisfaction with prompt after sales service.

The IDI Value Chain

1. Design & Innovation

We offer highly customized and industry specific chlorination systems for your chlorination needs. Each industry we serve, be it municipal water treatment or process water at thermal power plants, refineries, paper mills, pharmaceuticals or fertilizers, comes with its own set of design challenges which we have addressed in the 5000+ projects we have executed since our inception. We design our systems to ensure efficiency, safety and economy.

2. Manufacturing & Supply

Our efficient manufacturing facilities and our experienced shop floor team help give shape to our designs. By adherence to international standards and safety norms both in procurement and manufacturing, we guarantee world-class quality and finish for all our products. An independent quality team ensures quality at every stage.

3. Quality

We are an ISO 9001:2008 certified company and we follow quality process and procedures with interim inspections, final inspections and rigorous testing before shipping. Our systems have been successfully inspected and certified by various private and government testing and inspections bodies.

4. Erection & Commissioning

Probably the most important and crucial part of any project is the installation and commissioning. Having performed more than 5000+ projects, we have years of hands-on knowledge and skill when it comes to efficient, time-bound, and cost effective execution. Our team is equipped to handle all kinds of challenges faced at both local and international sites.

5. Training, maintenance & After Sales Support

We take pride in saying that we are one of the very few companies who provide prompt after sales service. Our qualified site engineers can offer training, maintenance and support services for all our chlorination systems. The operation of the system can either be taken care of by the client's staff or could be managed by our team as well. The client's operating staff are equipped with system and safety procedures through our intensive training programs. We offer annual maintenance contracts (AMC) to visit the site for regular check-ups and scheduled maintenance. We also undertake other service requirements like repair, replacement of specific parts or the whole system. We also train all personnel who are directly involved in the operation and maintenance of IDI chlorination systems.

PRODUCTS

Floor Mounted Chlorinator (Model: INDICHLOR - VF)

This is a floor mounted, vacuum operated, solution feed type chlorinator.

It incorporates chlorine pressure and vacuum gauges, efficient vacuum regulator, and an accurate flow meter, a precise flow control valve, an effective differential regulator, a reliable injector, a drain valve, a dependable double check non return valve.

- For capacities from 100 gms/hr to 300 kgs/hr
- Auto Flow Proportionate feature (Optional)
- MOC: FRP Cabinet, HDPE, PVC, PTFE, Hastalloy C internals
- Suitable for : Drinking water, Effluent treatment, Sewage treatment, Cooling water and Raw water treatment in power plants.



Floor Mounted Chlorinator



Wall Mounted Chlorinator

Wall Mounted Chlorinator (Model: INDICHLOR - VW)

This is a wall mounted, vacuum operated, solution feed type chlorinator. It incorporates chlorine pressure and vacuum gauges, vacuum regulator, and flow meter, a precise flow control valve, an injector, drain valve and a double check non return valve.

- For capacities from 100 gms/hr to 5kgs/hr
- MOC:FRP Cabinet, HDPE, PVC, PTFE, Hastalloy C internals
- Suitable for : Drinking water, effluent treatment and sewage treatment

Cylinder Mounted Chlorinator (Model: INDICHLOR - CM)

The Indichlor CM Cylinder mounted chlorinators are versatile and of high quality. They are low cost solution feed units, which are vacuum operated. The chlorinator is mounted on the cylinder or a ton container by means of a yoke clamp connection, which offers a leak proof joint. The chlorine flow meter, flow control valve, pressure relief valve, injector, drain valve and non return valve are part of the chlorinator.

- For capacities from 100gms/hr to 10 kgs/hr
- MOC: HDPE/ABS, PVC, PTFE, Hastalloy C internals
- Suitable for: Swimming pool, Drinking Water, smaller capacity Water treatment plant and Potable water treatment in power plant.



Cylinder Mounted Chlorinator

Operating Principle of Vacuum Chlorinator

Pressurized Water passes through the injector to produce vacuum. This vacuum opens the check valve and passes through the differential regulator and extends the vacuum to the vacuum regulator. The chlorine gas under pressure from the container passes through a filter where ferric chloride impurities are filtered and through the pressure reducing valve enters the vacuum regulator. The chlorine gas under vacuum passes through the vacuum line before getting mixed with the water in the injector, where it forms a chlorine solution. The chlorine solution is then discharged to the dosing point. The vacuum regulator is of an exclusive design, which opens only when the vacuum is generated by the injector and it closes automatically when the vacuum pressure goes below the specific limit. The glass tube rotameter indicates the flow of chlorine gas which can be adjusted with the help of the flow control valve provided on the chlorinator.

As per IS and Chlorine Institute, USA recommendations, it is not safe to manifold more than 4 tonners/cylinders. Hence, to meet the chlorine dosing requirements of 50 kgs/hr and above we recommend an electrically heated or steam heated evaporator. Using this evaporator we can draw liquid chlorine from a single chlorine ton container upto 200 kgs/hr approximately and convert it into gas chlorine in the evaporator, then take it to the chlorinator for dosing.

Electrically Heated Evaporator (Model: INDICHLOR - EHE)

The evaporator or vaporizer is used where chlorine requirement is large in quantity. We offer electrically heated evaporator up to a capacity of 300 kg/hr. The material used for the evaporator is of very good quality and is highly resistant to chlorine corrosion. The design of the evaporator and its accessories ensure safe and accurate operation. The evaporator is designed and fabricated to meet the standards of The Chlorine Institute Inc. The electrically heated evaporator incorporates inter locks for water level and temperature controls.

- For capacities from 50 kgs/hr to 300 kgs/hr
- Suitable for – Cooling Water & Raw Water gas chlorination in power plants, bigger capacity Water treatment and sewage treatment plants.



Electrically Heated Evaporator

Steam Heated Vaporizer (Model: INDICHLOR - SHV)

High capacity steam heated vaporizer up to a capacity of 1,250 kgs/hr has been developed for paper pulp and chemical industries. These are supplied with pressure reducing valve and rotameters.

- For capacities from 50 kgs/hr to 1250 kgs/hr
- Suitable for: Paper pulp, chemical and pharmaceutical industries



Steam Heated Vaporizer

Auto Changeover System

Whenever continuous uninterrupted chlorine supply is required, an auto change over system is used. When one tonner gets empty this system helps in switching over from an empty ton container to a full ton container automatically.

These come in 2 types:

Vacuum Auto Changeover Device (Model: INDICHLOR - VACD)

Chlorine vacuum auto changeover system consists of switch over module, which does not require power supply and is suitable upto 15kgs/hr system. The toggle type arrangement is operated when one of the tonners/cylinders become empty and vacuum is generated triggering the change over.



Electrical Auto Changeover Device



Vacuum Auto Changeover Device

Electrical Auto Changeover Device (Model: INDICHLOR - EACD)

There are operating and standby manifolds. When the operating manifold becomes empty, it will automatically switch over to the stand by tonners/cylinders and vice versa. This system has been provided with pressure switches, motorized valves and an electrical panel.

SAFETY SYSTEMS



Leak Absorption System

Chlorine gas is hazardous and its leakage can cause danger to life. So any facility that uses chlorine must be provided with chlorine gas scrubber system. This system is switched on automatically on detection of chlorine leakage. The system includes a vertical packed tower, caustic storage and recirculation tank, recirculating pump, exhaust blower, valves, piping, duct work and control panel.

SAFETY ACCESSORIES

Chlorine Leak Detector

The chlorine gas detection system consists of a control unit, a sensor and a hooter. The sensor is mounted remotely, senses the gas present in the room, which is indicated digitally on the panel. Apart from the audio visual alarm provided in the panel, there is a potential free contact which can be used to trigger the remote alarm or actuate the exhaust fan/leak absorption system. Leak detectors are available with single or multiple sensors.



Chlorine Leak Detector

Emergency Repair Kit

Our kit has devices to stop leakage at the valve, inspection/fusible plug and from the wall of the chlorine container. The kit contains a canister type gas mask, protective clothing, necessary tools and tackles, gaskets, ammonia torch etc. A step-by-step instruction booklet is also provided with the kit. We manufacture both Kit A for small capacity cylinders & Kit B for upto 900 kg ton containers.



Emergency Repair Kit

Self contained Air Breathing Apparatus

Self - Contained Air Breathing apparatus are very essential for heavy toxic leaks. Manufactured as per IS: 10245, Part-II. Available in 30 minutes or 45 minutes duration.



Air Breathing Apparatus

Canister Gas Mask

The Gas Mask with Canister is designed to enable the wearer to breathe fresh air in a contaminated atmosphere. These are applicable for small leaks only.



Canister Gas Mask

Portable Air Compressor

This is a high pressure air compressor which can generate 200/300 bar. It is used to fill the air cylinders of the breathing apparatus.



Portable Air Compressor

Emergency Eye Wash and Shower

Emergency Eyewash and Shower (Hand Operated or Pedal Operated - Combination units) is required to protect eye, face & skin from chlorine contacts. The deluge of water dilutes and washes away harmful material, thereby completely eradicating chances of losing vision and burn injuries. It is ISI marked and conforms to IS : 10592, ANSI Z 358.1-2009.



Emergency Eye Wash & Shower

Instant Air Resuscitator

This consists of a mask and a large hand-squeezed plastic bulb using ambient air to resume artificial respiration when breathing ceases.

ACCESSORIES



Lifting Beam

The lifting beam is very essential for safe handling of chlorine ton container. This can also be fitted with a dynamometer (load indicator) to weigh the ton container.



Auxiliary Container Valve

An additional valve assembly mounted with a yoke on the container valve to keep the container valve damage free.



Gas Filter

It is mounted on the gas phase which eliminates the carry over of ferric chloride and other impurities.

Other accessories:

- Residual chlorine analyser
- Residual test kit
- Motorized ball valve
- Chlorine valve
- Pressure gauge (Diaphragm type)
- Basket strainer



Trunnion Roller Supports

These help in the proper placement of tonners and to rotate the ton container easily for positioning the gas valve.



Flexible Copper Connector

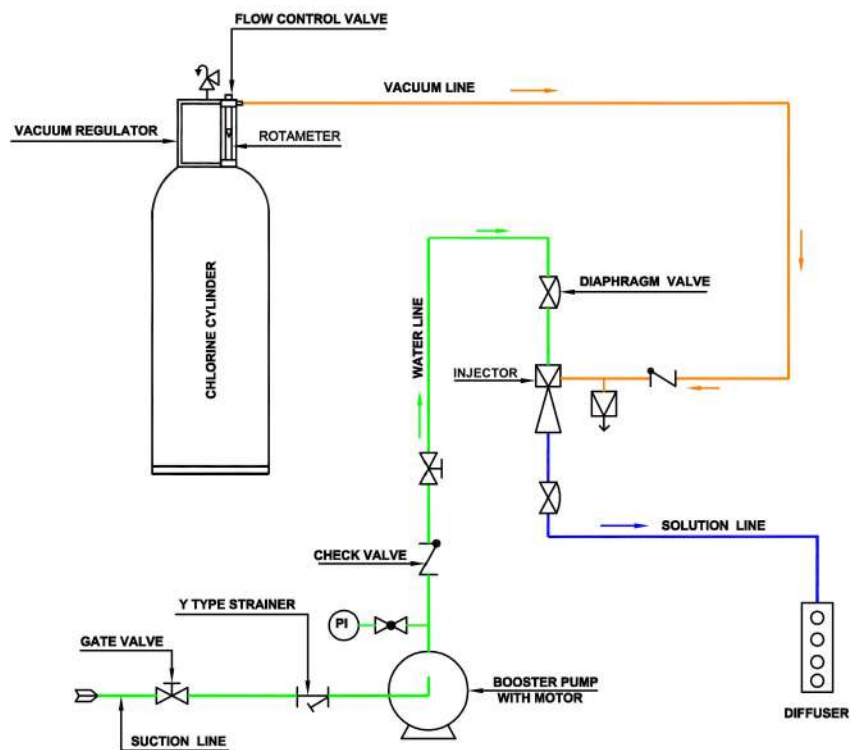
This connector offers positive connection between the tonner and the manifold



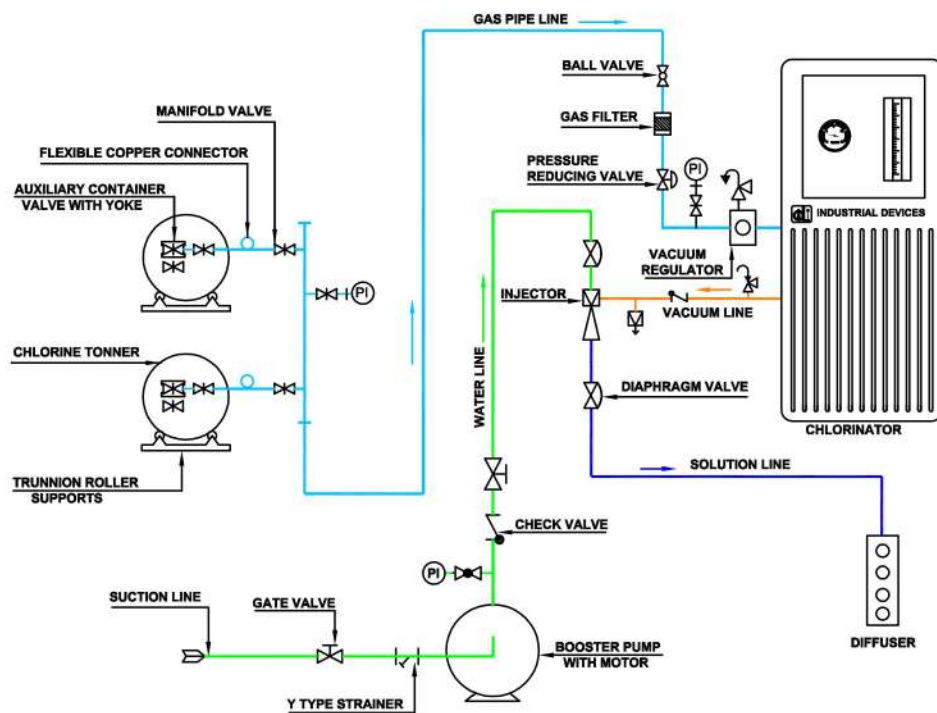
Pressure Reducing Valve

This valve is capable of reducing the pressure of chlorine gas to a manageable limit of 2 to 2.5 kg/cm². The low pressure prevents the re-liquefaction of chlorine gas in the system.

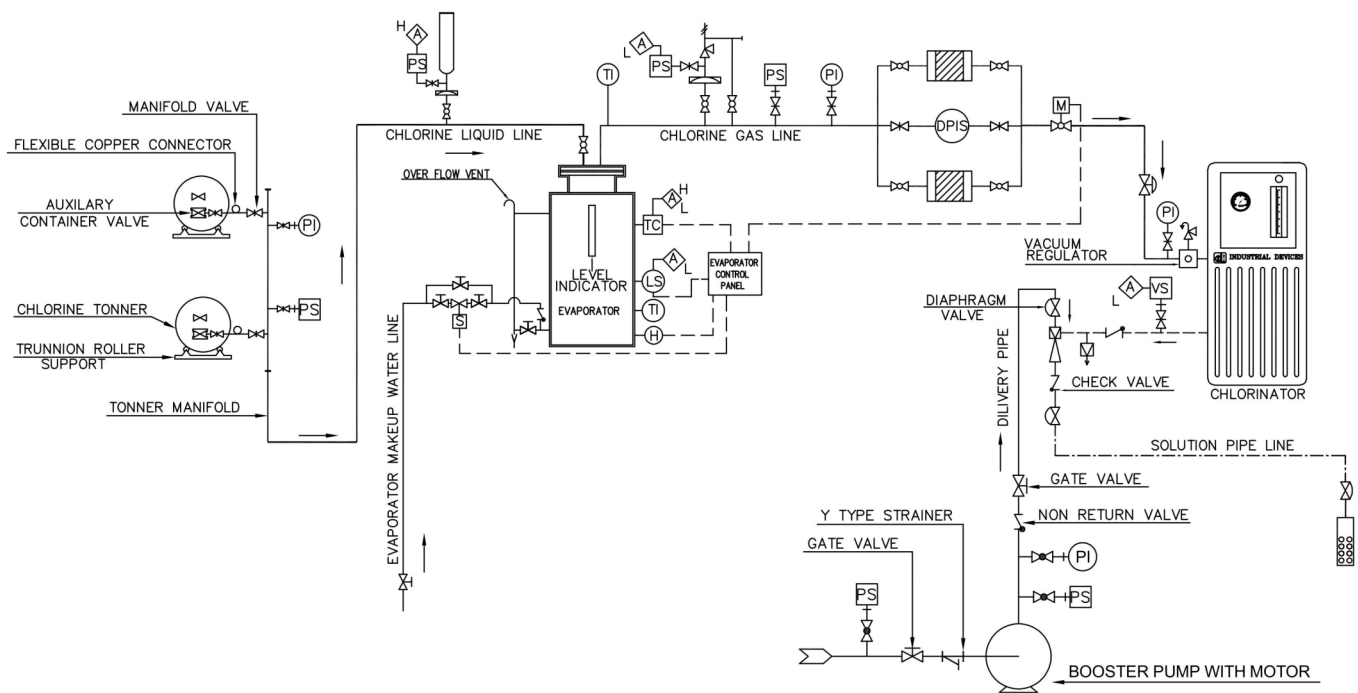
TYPICAL SYSTEM DESIGN



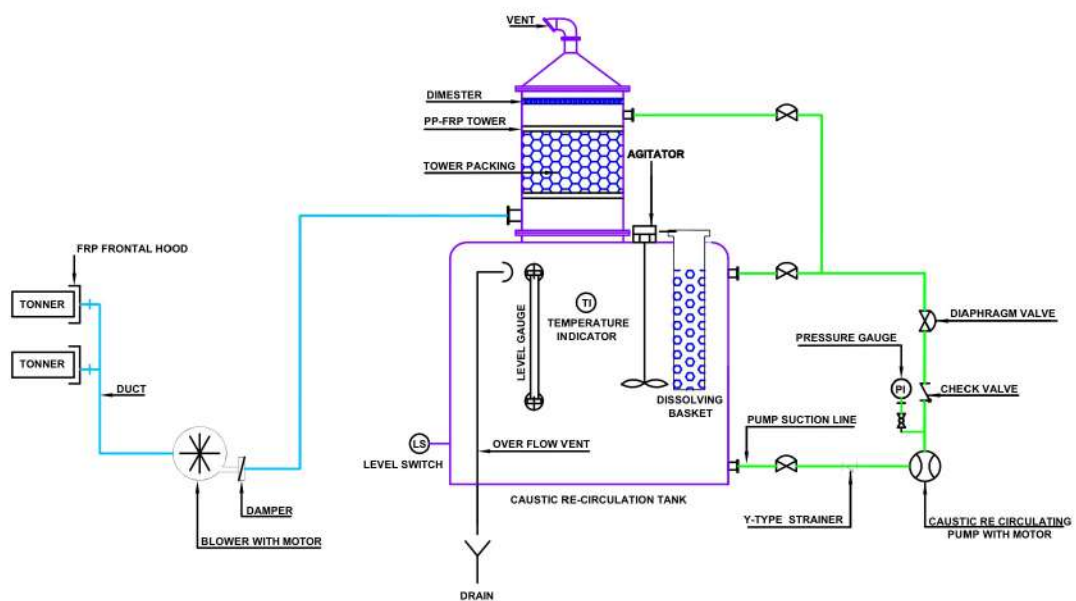
CYLINDER MOUNTED CHLORINATION SYSTEM



CHLORINATION SYSTEM WITHOUT EVAPORATOR



CHLORINATION SYSTEM WITH EVAPORATOR



LEAK ABSORPTION SYSTEM

WE MANUFACTURE BOTH FORCED DRAFT AND INDUCED DRAFT CHLORINE LEAK ABSORPTION SYSTEM

SOME INSTALLATIONS



100 Kgs/hr CW Chlorination Plant
at TS I Expansion TPS Neyveli



200 Kgs/hr CW Chlorination system,
Mettur TPS TANGEDCO



Pre and Post Chlorination System at 205
MLD WTP HMWSSB Hyderabad



Pre and Post Chlorination System at 410
MLD WTP HMWSSB Hyderabad



50 Kgs/hr CW Chlorination System at
KTPS A Stn. Kothagudam Telangana Power



80 Kgs/hr CW Chlorination System at
RTPP Rayalseema APGENCO

KEY CLIENTS

Sectors	Capacity
Thermal Power Plants	
1. 2x300 MW TPS , Lanco Amarkantak , Korba, Chhattisgarh.	3 nos. of 200 kgs/hr CW Chlorination System with Leak Absorption system.
2. 600 MW Stage-III, Mettur TPS, TANGEDCO, Tamil Nadu	2 nos. of 200 kgs/hr CW Chlorination System with Leak Absorption system.
3. 2x500 MW Bhusawal TPS, MAHAGENCO, Maharashtra	3 nos. of 150 kgs/hr CW Chlorination System with Leak Absorption system
4. TS-1 Expansion, Neyveli Lignite Corporation, Neyveli, Tamil Nadu	3 nos. of 100 kgs/hr Chlorination System with Leak Absorption and Chemical Dosing System
5. 2x500 MW Chandrapur STPS Expansion, MAHAGENCO, Chandrapur, Maharashtra.	3 nos. of 150 kgs/hr CW Chlorination System with Leak Absorption system
Water Treatment Plants	
6. 205 MLD WTP at Kodandapur Stage-I, Phase-I (HMWSSB), Hyderabad (Through M/s. Geo Miller & Co. Pvt. Ltd., New Delhi)	2 nos. of 60 kgs/hr Pre Chlorination System with Evaporator and 2 Nos. of 20 kgs/hr Post Chlorination System.
7. 410 MLD WTP at Kodandapur Stage-I, Phase-II, MWSSB, Hyderabad (Through M/s. Geo Miller & Co. Pvt. Ltd., New Delhi)	4 nos. of 60 kgs/hr Pre Chlorination System with Evaporator and 4 Nos. of 20 kgs/hr Post Chlorination System.
8. 74 MLD WTP at Nokha Daiya, Bikaner (Through M/s. Pratibha Industries Limited, Mumbai)	2 nos. of 50 kgs/hr Pre Chlorination System and 2 Nos. of 30 kgs/hr Post Chlorination System.
9. 410 MLD WTP at Kodandapur, HMWSSB, Hyderabad (Through M/s. Venky Infra)	4 nos. of 60 kgs/hr Pre Chlorination System with Evaporator and 4 Nos. of 20 kgs/hr Post Chlorination System.
10. 200 MLD WTP Warje, Pune (Through M/s. Geo Miller & Co. Pvt. Ltd., New Delhi)	4 nos. of 40 kgs/hr Chlorination System
Sewage Treatment Plants	
11. 90 MLD STP at Belandur, BWSSB, Bangalore (Through M/s. VA Tech Wabag Ltd., Chennai)	2 nos. of 40 kgs/hr Chlorination System and Leak Absorption system.
12. 125 MLD STP at Madurai (Through M/s. Ramky Infrastructure Ltd.)	2 nos. of 30 kgs/hr Chlorination System
13. 120 MLD STP at Koyambedu, CMWSSB, Chennai (Through M/s. S N Gharepure)	2 nos. of 30 kgs/hr Chlorination System
Paper and Chemical Plants	
14. M/s. IICP, Egypt	3 nos. of 1250 kgs/hr Steam Heated Vaporizer system
15. M/s. Demaco Polymers Ltd. , Haryana	1 no. of 1000 kgs/hr Steam Heated Vaporizer system
16. M/s. High Polymers Ltd., Haryana & Punjab	2 nos. of 700 kgs/hr Steam Heated Vaporizer system
Overseas Projects	
17. M/s. General Corp of water and sewerage, Baghdad - Iraq	500 nos. of Vacuum Type Chlorinators of capacities ranging from 500 gms/hr to 20 kgs/hr
18. M/s. Industrial Chemical Park SAE, Egypt	3 nos. of 1250 kgs/hr Steam Heated Chlorine Vaporizer
20. M/s. Power Engineering World Ltd. Co., Jordan	6 nos. of 10 kgs/hr Chlorination system for Baghdad Water Authority.
19. M/s. Blue Ocean Co. Ltd., Tehran	100 nos. of 2 kgs/hr Cylinder Mounted Chlorinator.
21. M/s. Roytex SA for a power project at Guatemala	1 no. of 15 kg/hr Chlorination and Leak Absorption System
22. M/s. KBL (Gondwana Engineers Pvt. Ltd.), Lesotho (Near South Africa)	1 no. of 5 kg/hr Chlorination System
We are OEM suppliers to all major Water Treatment and EPC companies.	

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